# Harshita Lila

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## EDUCATION

**B.E. (Electronics and Computer Engineering)** 2021 – 2025
*MBM University, Jodhpur*
CGPA: 7.45

**12th** 2020
*St. Anne's Sr. Sec. School, Jodhpur*
Percentage: 84.2

**10th** 2018
*St. Anne's Sr. Sec. School, Jodhpur*
Percentage: 86.4

## INTERNSHIPS

**DRDO, Jodhpur** 2023
- Worked as a research trainee at DRDO, Jodhpur.
- Studied radio communication and implemented a temperature monitoring system to facilitate remote monitoring of system health.

## TECHNICAL SKILLS

- Programming Languages: C, C++, Python
- Systems Programming
- Linux
- SQL (MySQL, SQL Server Management Studio)
- Deep Learning
- Image Processing (OpenCV/MATLAB)
- Microcontrollers: Arduino, Raspberry Pi, ESP32
- Robotics and Control
- Automation
- Digital Logic

## PROJECTS

**McMini: DPOR based model checker** 2025
- Added heuristic for livelock detection.
- Added LLVM instrumentation plugin to enable detection of reads and writes to global variables, thus enabling detection of data races within an executable.
- Implemented command line flags to enhance usability.

***Systems Programming, Model Checking, LLVM instrumentation***

**Autonomous Damage Control & Monitoring with Real Time Tracking** 2025
- An autonomous robot designed with the view of delivering essential utilities in disaster affected areas.
- A watchtower equipped with a high resolution camera oversees the arena and detects any damage using a neural network model.
- In case of multiple instances of such events, a pre decided priority order is used to determine the optimal path that needs to be followed.
- Control signals are passed to the robot wirelessly which then uses line following to traverse the arena.
- Live location of the robot is plotted on a map to monitor its movement.

***Robotics and control, Deep Learning, Path planning, Wireless communication, Image Processing***

**Transparent Checkpointing** 2024
- Implemented transparent checkpointing using C.
- The program handles SIGUSR2 to create a checkpoint and in case of incorrect termination of process, restarts it from the saved checkpoint transparently.

***Advanced C, Systems programming, Checkpointing***

**WiSh: Linux shell** 2024
- Implemented an interactive shell in C language, supporting all the basic linux commands and offers **cd, path** and **exit** as built-in commands.
- Support for I/O redirection has also been added.

***C, Systems Programming, Command parsing, Processes, Process control, System calls***

**Two Link Robotic Arm Model for playing 5x5 Tic-Tac-Toe with Human** 2023
- Built a two link robotic arm model capable of playing optimal tic-tac-toe against a human opponent.
- The robot captures real time position of the game board and utilizes the minimax algorithm to calculate the next best move.
- Alpha beta pruning is employed to further optimize the process.
- A magnetic gripper is used to pick up checkers from a rack and place on the game board.

***Robotics and control, kinematics, image processing, wireless communication***

**Maze Solver Robot** 2022

- Implemented a maze solver robot that utilises line following to explore the maze during a 'dry run' and then traverses through it using the shortest path.

***Microcontrollers, Robotics, Maze solving algorithm***

**Database Management System** 2019

- Built a database management system in C++ that can create tables with upto **10** fields of either int or string type and offers support for basic operations like ***search, update, add, delete and sort records***.
- The DBMS takes string type commands as input, parses them and performs the required operation.

***C++, Database, File management, Command parsing***

## COMPETITIONS & AWARDS

**Student Achievers Award 2025**
Received the Student Achievers Award for the batch of 2025, awarded annually to the top 4 students of MBM University who have shown exemplary achievements in all domains (academics, art & culture, sports, community service etc).

**E-Yantra Robotics Competition 2023**
Participated in E-Yantra Robotics competition and successfully completed Stage 1 of the 6 month long competition organised by IIT Bombay.

**Encarta 2022**
Won 1st prize in line follower robot competition held during Encarta 2022, annual technical fest organised by MBM University.

**Google Code in 2018**
Participated in Google Code-in, 2018 and completed **2 tasks** set by Haiku, an open source Operating System.

## ROLES OF RESPONSIBILTY

**Student coordinator, Embedded Systems & Robotics Club** 2022 – 2025

- Served as the student coordinator for Embedded Systems and Robotics Club, MBM University, Jodhpur for 3 consecutive terms.
- Organised **three 45-day** robotics workshops.
- Mentored **500+** students.

**Event coordinator, TechKriti 2023 (MBM University)** 2023

- Served as an event coordinator and helped organise the annual technical fest of MBM University, Jodhpur which witnessed a footfall of over **2000** people.
- Ensured smooth working of a team of over **50 people**.
- Curated quality content to be printed in brochures and other advertising material.

**Wiki Lead, GDSC (MBM University)** 2022 – 2023
Served as the wiki lead for Google Developer Students Club, MBM University, Jodhpur and curated quality technical content for the student community.

## SOFT SKILLS

**Leadership**

**Management**

**Teamwork**

**Communication**